

Additional file 1

Fixed-margin prediction of cell-level RNA–surface-protein dependence across cohorts

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1 Coupling coordinates

1.1 Definition

Let $P \in (0, 1)^{r \times s}$ be a joint probability table, and let $B_r \in \mathbb{R}^{r \times (r-1)}$ and $B_s \in \mathbb{R}^{s \times (s-1)}$ have orthonormal columns spanning the complements of the all-ones vectors. The population coupling field is

$$\Theta(P) = B_r^\top \log(P) B_s. \quad (1)$$

We use Helmert bases. Other orthonormal contrast bases rotate the $(r-1)(s-1)$ coordinates without changing Frobenius norms or pairwise Frobenius distances.

Proposition S1 (Marginal invariance and independence). *For positive vectors $a \in \mathbb{R}^r$ and $b \in \mathbb{R}^s$, define*

$$P' = \frac{\text{diag}(a)P \text{diag}(b)}{\sum_{u,v} a_u P_{uv} b_v}.$$

Then $\Theta(P') = \Theta(P)$. Moreover, $\Theta(P) = 0$ if and only if P is an independent joint law.

Proof. Writing c for the normalizing constant gives

$$\log P' = \log P + (\log a) \mathbf{1}_s^\top + \mathbf{1}_r (\log b)^\top - \log(c) \mathbf{1}_r \mathbf{1}_s^\top.$$

The added terms are annihilated on the left by $B_r^\top$ or on the right by B_s , proving invariance. Factorization $P_{uv} = p_u q_v$ makes $\log P$ a sum of row and column terms, so $\Theta(P) = 0$. Conversely, $\Theta(P) = 0$ removes the interaction component in the orthogonal decomposition into row and column terms. Hence $\log P_{uv} = \alpha_u + \beta_v$, and P factorizes after normalization. $\square$

2 Binary fixed-margin estimator

The primary estimator conditions binary RNA-protein tables on their observed margins, which eliminates nuisance marginal parameters. Predictions use the recipient margins without a pseudocount.

2.1 Binary scaling identity

For

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}, \quad B_2 = 2^{-1/2} (1, -1)^\top,$$

Eq. (1) is scalar and satisfies

$$\Theta(P) = \frac{1}{2} \log \frac{p_{00} p_{11}}{p_{01} p_{10}}. \quad (2)$$

The conventional log odds ratio is therefore $\theta = 2\Theta(P)$. Estimation uses θ ; reported field coordinates use the Helmert scale in Eq. (2).

2.2 Fixed-margin likelihood

For donor d and ordered RNA-protein pair e , write the 2×2 table as

$$N_{de} = \begin{pmatrix} A_{de} & r_{0,de} - A_{de} \\ c_{0,de} - A_{de} & n_{de} - r_{0,de} - c_{0,de} + A_{de} \end{pmatrix},$$

where $r_{0,de}$ and $c_{0,de}$ are the first row and column totals and $c_{1,de} = n_{de} - c_{0,de}$. Conditional on these margins,

$$\Pr_{\theta_{de}}(A = a \mid r_0, c_0, n) = \frac{\binom{c_0}{a} \binom{c_1}{r_0 - a} \exp(\theta_{de} a)}{Z_{de}(\theta_{de})}, \quad Z_{de}(\theta) = \sum_{a \in \mathcal{A}_{de}} \binom{c_0}{a} \binom{c_1}{r_0 - a} \exp(\theta a), \quad (3)$$

on the feasible integer support $\mathcal{A}_{de}$. Omitting a constant independent of θ , the negative log likelihood is

$$\ell_{de}(\theta) = \log Z_{de}(\theta) - A_{de} \theta, \quad \ell'_{de}(\theta) = \mathbb{E}_\theta[A] - A_{de}, \quad \ell''_{de}(\theta) = \text{Var}_\theta(A). \quad (4)$$

An informative table has at least two values in $\mathcal{A}_{de}$, making $\ell''_{de} > 0$ and the conditional odds ratio identifiable.

Hierarchical conditional and random-effects log-linear models share odds-ratio information across 2×2 strata [1, 2, 3]. We estimate donor-specific and population log odds jointly across the entity panel, then predict recipient tables at their observed margins.

2.3 Hierarchical transfer objective

Let D denote source donors, E the ordered marker pairs, θ_{de} a donor-specific log odds ratio, and μ_e the transferable population log odds. Two marker hypergraphs induce normalized Laplacians L_R and L_A for the RNA and protein axes. Their product-space penalty is

$$L = L_R \otimes I + I \otimes L_A. \quad (5)$$

The estimator minimizes the penalized composite conditional objective

$$Q(\theta, \mu) = \frac{1}{D} \sum_{d=1}^D \sum_{e=1}^E \ell_{de}(\theta_{de}) + \frac{\gamma}{2} \sum_{d,e} (\theta_{de} - \mu_e)^2 + \frac{\rho}{2} \|\mu\|_2^2 + \frac{\tau}{2} \mu^\top L \mu. \quad (6)$$

The likelihood sum is composite because entity pairs within a donor share cells and markers. We express the penalty grid through dimensionless inputs

$$\gamma = \eta s_\eta / D, \quad \rho = \lambda_0 s_\mu, \quad \tau = \lambda_G s_\mu,$$

where s_η is the median null Fisher information across informative donor-entity tables and s_μ is the median information after donor averaging. This scaling makes the grid invariant to duplication of the complete donor panel.

Its gradients are

$$\frac{\partial Q}{\partial \theta_{de}} = \frac{\mathbb{E}_{\theta_{de}}[A] - A_{de}}{D} + \gamma(\theta_{de} - \mu_e), \quad (7)$$

$$\nabla_\mu Q = -\gamma \sum_d (\theta_d - \mu) + \rho \mu + \tau L \mu. \quad (8)$$

The Hessian has diagonal θ block $\text{diag}\{\text{Var}_{\theta_{de}}(A)/D + \gamma\}$, θ - μ entries $-\gamma$, and μ block $D\gamma I + \rho I + \tau L$. Its quadratic form is

$$\sum_{d,e} \frac{\text{Var}_{\theta_{de}}(A)}{D} u_{de}^2 + \gamma \sum_{d,e} (u_{de} - v_e)^2 + \rho \|v\|_2^2 + \tau v^\top L v. \quad (9)$$

Proposition S2 (Finite identifiable fit). *If $\gamma > 0$, $\rho > 0$, and $\tau \geq 0$, Eq. (6) has a unique finite minimizer. For every informative donor-entity table, its unpenalized conditional parameter is identifiable.*

Proof. Each likelihood term is convex by Eq. (4), and the product Laplacian is positive semidefinite. Equality to zero in Eq. (9) requires $u_{de} = v_e$ through the γ term and $v = 0$ through the ρ term, hence $u = 0$. The objective is strictly convex and coercive. The final statement follows from $\ell''_{de} > 0$ on every informative support. $\square$

Newton updates eliminate donor-specific parameters by the Schur complement. The population diagonal then receives

$$\sum_d \frac{\gamma \{\text{Var}_{\theta_{de}}(A)/D\}}{\gamma + \text{Var}_{\theta_{de}}(A)/D}$$

in addition to $\rho I + \tau L$. Backtracking enforces descent, and fits that exceed prespecified gradient-norm or condition-number thresholds are rejected.

For a recipient donor, Eq. (3) with log odds $\alpha \hat{\mu}_e$ and the recipient margins yields a conditional-expectation prediction that preserves both margins. Development donors determine α and all penalties.

2.4 Source-only graphs and tuning grids

Graphs used only samples available at the corresponding selection stage. For Stephenson, GSE239452, and COMBAT, standardized marker-prevalence profiles across source samples defined Euclidean k -nearest-neighbor graphs for each assay; the union of directed neighbor relations formed undirected two-vertex hyperedges. BMBC used prevalence across source donor-by-lineage strata, with each marker and its k nearest neighbors forming one unweighted hyperedge. The resulting normalized RNA and ADT hypergraph Laplacians entered Eq. (5). Cross-validation rebuilt both graphs within each calibration fold, and the final fit used all permitted development samples.

Table 1 gives the complete binary grids. Redundant values of k were omitted when $\lambda_G = 0$. The statistic comparator crossed Pearson and deviance statistics, raw and conditional-null-centered forms, and the same transport grid; GSE239452 evaluated raw statistics only.

2.5 Transfer risk and recipient drift

To separate source estimation error from recipient variation, fix informative margins $M = (r_0, r_1, c_0, c_1)$ with total N and endpoints

$$L_M = \max(0, r_0 + c_0 - N), \quad U_M = \min(r_0, c_0)$$

for the feasible upper-left count. Write Eq. (3) as

$$p_\theta(a \mid M) = h_M(a) \exp\{\theta a - \psi_M(\theta)\}, \quad I_M(\theta) = \psi_M''(\theta) = \text{Var}_\theta(A \mid M).$$

Table 1: Binary estimator grids used for the reported analyses.

Analysis	Neighbors k	Heterogeneity η	Ridge λ_0	Graph λ_G	Transport α
Stephenson	1, 2	0.1, 1, 10	0.01, 0.1	0, 0.1, 1	0.5, 0.75, 1, 1.25
GSE239452	1, 2	0.1, 1, 10	0.01, 0.1	0, 0.1, 0.3, 1	0.5, 0.75, 1, 1.25
COMBAT	1, 2	0.1, 1, 10	0.01, 0.1	0, 0.1, 0.3, 1	0.5, 0.75, 1, 1.25
development					
BMMC	1, 2, 3	0.1, 1, 10	0.01, 0.1	0, 0.1, 0.3, 1	0.5, 0.75, 1, 1.25
development					

Proposition S3 (Fixed-margin transfer risk). *Let $\mathcal{S}$ contain all source information, and suppose a recipient log odds satisfies*

$$\theta_q = \mu + \xi_q, \quad \mathbb{E}(\xi_q \mid \mathcal{S}, M) = 0, \quad \mathbb{E}(\xi_q^2 \mid \mathcal{S}, M) = \sigma_M^2.$$

The source estimate $\hat{\mu}$ and multiplier α are $\mathcal{S}$ -measurable, and the recipient prediction uses $\hat{\theta} = \alpha\hat{\mu}$. Assume that θ_q and θ lie almost surely in a compact interval $\mathcal{T}$. Define

$$\underline{I}_M = \inf_{\theta \in \mathcal{T}} I_M(\theta), \quad \bar{I}_M = \sup_{\theta \in \mathcal{T}} I_M(\theta).$$

Then the conditional excess log loss

$$\mathcal{R}_M = \mathbb{E}[\text{KL}\{p_{\theta_q}(\cdot \mid M) \parallel p_{\hat{\theta}}(\cdot \mid M)\} \mid \mathcal{S}, M]$$

satisfies

$$\frac{\underline{I}_M}{2} \{(\alpha\hat{\mu} - \mu)^2 + \sigma_M^2\} \leq \mathcal{R}_M \leq \frac{\bar{I}_M}{2} \{(\alpha\hat{\mu} - \mu)^2 + \sigma_M^2\}. \quad (10)$$

Moreover, $I_M(\theta) \leq (U_M - L_M)^2/4$ for every finite θ , so

$$\mathcal{R}_M \leq \frac{(U_M - L_M)^2}{8} \{(\alpha\hat{\mu} - \mu)^2 + \sigma_M^2\}. \quad (11)$$

Divide these bounds by N for excess log loss per cell.

Proof. For canonical parameters θ and η ,

$$\text{KL}(p_\theta \parallel p_\eta) = \psi_M(\eta) - \psi_M(\theta) - \psi'_M(\theta)(\eta - \theta).$$

Taylor's theorem with integral remainder gives

$$\text{KL}(p_\theta \parallel p_\eta) = (\eta - \theta)^2 \int_0^1 (1-t) I_M\{\theta + t(\eta - \theta)\} dt.$$

Bounding the information along the segment from θ to η , substituting $\theta = \mu + \xi_q$ and $\eta = \alpha\hat{\mu}$, and taking the conditional expectation prove Eq. (10). Popoviciu's inequality gives $I_M(\theta) = \text{Var}_\theta(A \mid M) \leq (U_M - L_M)^2/4$, which proves Eq. (11). For informative margins, I_M is continuous and positive on the compact interval $\mathcal{T}$, hence $\underline{I}_M > 0$. $\square$

If α is fixed across source repetitions and $b = \mathbb{E}(\hat{\mu}) - \mu$, the reducible term in Eq. (10) decomposes as

$$\mathbb{E}(\alpha\hat{\mu} - \mu)^2 = \alpha^2 \text{Var}(\hat{\mu}) + \{\alpha b + (\alpha - 1)\mu\}^2. \quad (12)$$

As source estimation becomes consistent and α approaches one, σ_M^2 remains; fixed-margin Fisher information converts this recipient drift into an irreducible conditional-risk floor.

2.6 Decision rule for table prediction

The conditional model defines a distribution over integer tables with the recipient margins. The prediction used in the paper is its mean table.

Proposition S4 (Conditional mean minimizes expected deviance). *Let N be a random contingency table with fixed margins M and total n , and let $\bar{N} = \mathbb{E}(N \mid M)$. For any nonnegative table Q with margins M and $Q_{ij} > 0$ whenever $\bar{N}_{ij} > 0$,*

$$\mathbb{E}\{D(N, Q) \mid M\} - \mathbb{E}\{D(N, \bar{N}) \mid M\} = \frac{2}{n} \sum_{i,j: \bar{N}_{ij} > 0} \bar{N}_{ij} \log \frac{\bar{N}_{ij}}{Q_{ij}} \geq 0. \quad (13)$$

The conditional mean is the unique minimizer on cells with positive conditional mean.

Proof. For any admissible Q ,

$$\mathbb{E}\{D(N, Q) \mid M\} = \frac{2}{n} \mathbb{E} \left(\sum_{i,j: N_{ij} > 0} N_{ij} \log N_{ij} \mid M \right) - \frac{2}{n} \sum_{i,j} \bar{N}_{ij} \log Q_{ij}.$$

The first term is independent of Q . Subtracting the same expression evaluated at $\bar{N}$ gives Eq. (13). Because Q/n and $\bar{N}/n$ are probability tables, the right side is twice their Kullback–Leibler divergence. It is nonnegative and vanishes only when $Q = \bar{N}$ on the support of $\bar{N}$. Every table in the conditional support has margins M , so linearity of expectation gives the same margins to $\bar{N}$. $\square$

2.7 Matched signed-statistic transfer

The matched statistic-transfer comparator uses the same binary tables, donors, splits, recipient margins, selection stage, and deviance-per-cell outcome. Stephenson pools donor statistics with the Paule–Mandel random-effects estimator [4]. For determinant $D_N = N_{00}N_{11} - N_{01}N_{10}$, the signed root Pearson statistic is

$$R_P = \text{sign}(D_N) \sqrt{X^2} = D_N \sqrt{\frac{n}{r_0 r_1 c_0 c_1}}. \quad (14)$$

The signed root likelihood-ratio deviance statistic is

$$R_G = \text{sign}(D_N) \left\{ 2 \sum_{i,j: N_{ij} > 0} N_{ij} \log(N_{ij}/E_{ij}) \right\}^{1/2}, \quad E_{ij} = r_i c_j / n. \quad (15)$$

Statistics are divided by $\sqrt{n}$ before donor pooling, multiplied by α , and restored to the recipient $\sqrt{n}$ scale before fixed-margin reconstruction. Stephenson used donor-equal Paule-Mandel pooling and selected the statistic, its raw or conditional-null-centered form, and α on the pilot. GSE239452 used equal donor means and selected a raw statistic and α by calibration cross-validation. COMBAT and BMMC used equal donor means and evaluated both centering rules. The hierarchical method transported conditional log odds and reconstructed the noncentral-hypergeometric expectation.

2.8 Fitted interaction baselines

For each entity e , the common-effect conditional maximum-likelihood estimate solves

$$\sum_d \mathbb{E}_{\theta_e}[A_{de} \mid m_{de}] = \sum_d A_{de}$$

The implementation returns the corresponding extended estimate at a support boundary and otherwise requires an absolute score below 10^{-8} . This estimator has no donor deviations, ridge penalty, or graph penalty.

The pooled-table baseline first sums the source-donor tables for each entity and computes the interaction coefficient from the saturated model

$$\log \mathbb{E}[N_{ij}] = \beta_0 + \beta_i^R + \beta_j^C + \beta^I \mathbb{I}\{i = 1, j = 1\}.$$

For a 2×2 table, β^I is the pooled sample log odds ratio, which differs from the stratified conditional estimate when donor margins vary. In the original post hoc comparison, both fitted coefficients select α on development donors and reconstruct held tables with Eq. (3). The pooled method is therefore pooled-table log odds with conditional reconstruction, not a Poisson plug-in prediction.

A separate post hoc comparator fixes the same pooled saturated-model interaction and profiles recipient row and column parameters. This gives the unique positive continuous table with the recipient margins and transported odds ratio, without Eq. (3). The original calibration and pilot donors select α before the interaction is refitted on all development donors.

2.9 Ground-truth estimator simulation

The factorial simulation crossed donor heterogeneity with recipient-margin shift. Each of 128 seeded replicates contained 12 source donors, 12 recipients, 16 independent binary entity tables per donor, and 128 observations per table. Counts followed the Fisher noncentral-hypergeometric distribution at known population log odds. Source donor log odds either equaled the population value or had independent normal deviations with standard deviation 1.2. Recipient margins were source-like or shifted toward opposing row and column prevalences, using common random streams across margin conditions.

Across scenarios, the hierarchical estimator used heterogeneity penalty 0.1, ridge penalty 0.01, graph penalty zero, and $\alpha = 1$. The common-effect estimator fitted one unpenalized conditional log odds across source donors; raw and conditional-null-centered signed-root deviance statistics also used $\alpha = 1$. Recipient outcomes entered only the final deviance calculation. Intervals used 20,000 paired bootstrap draws over replicates.

Table 2: Ground-truth simulation: mean recipient multinomial deviance per cell. Parentheses give paired 95% bootstrap intervals; lower is better.

Donor effects	Recipient margins	Hierarchical	Common conditional	Raw statistic	Hierarchical comparison
Heterogeneous	source-like	0.014931	0.015212	0.015508	1.85% below common (1.52 to 2.18%); 3.72% below statistic transfer (3.28 to 4.15%)
Heterogeneous	shifted	0.011591	0.011714	0.033587	1.05% below common (0.84 to 1.26%); 65.49% below statistic transfer (63.02 to 67.72%)
Homogeneous	source-like	0.008752	0.008660	0.008688	1.06% above common (0.61 to 1.51%); 0.75% above statistic transfer (0.21 to 1.28%)
Homogeneous	shifted	0.008674	0.008640	0.029427	0.40% above common (0.13 to 0.67%); 70.52% below statistic transfer (69.21 to 71.74%)

2.10 Transfer-risk stress test

The stress test fixed the source field at its known value and crossed bounded mean-zero recipient drift, transport multipliers $\{0.5, 0.75, 1, 1.25\}$, drift standard deviations $\{0, 0.25, 0.5, 1\}$, and three informative margins. Finite-support summation gave the expected conditional KL risk, numerical quadrature evaluated its curvature-integral form, and 10^6 seeded conditional draws per drift state gave an independent Monte Carlo estimate.

All 48/48 conditions satisfied the Fisher-information and finite-support envelopes. Exact KL and its curvature integral differed by at most 1.91×10^{-14} . Monte Carlo intervals covered the exact risk in 48/48 conditions, with maximum absolute $z = 1.833$. At $\alpha = 1$, increasing the drift standard deviation from 0 to 1 raised count-level expected log loss from 0 to 2.835 for balanced margins, 1.504 for row-sparse margins, and 1.322 for overlap-shifted margins; each table contained 96 observations.

3 Secondary multistate methods

3.1 Permutation-centered fields

For linked state vectors $(U_i, V_i)_{i=1}^n$, let $h_\alpha(U, V)$ denote Eq. (1) applied to the joint count table after adding pseudocount α . With B independent permutations of the V labels, the centered field is

$$\hat{\Theta} = h_\alpha(U, V) - B^{-1} \sum_{b=1}^B h_\alpha(U, V^{(b)}). \quad (16)$$

Under the fixed-margin link null, observed and permuted matchings are identically distributed conditional on both label multisets, so Eq. (16) has conditional mean zero. If S^2 is the coordinatewise sample variance of the B reference fields, $(1 + 1/B)S^2$ is an unbiased estimator of the variance of an observed-minus-reference field. Multistate analyses use $B = 64$, $\alpha = 0.5$, and one independent destroyed-link permutation.

For perturbation g and control 0 under challenge 1 and vehicle 0, the factorial field is $(\Theta_{g1} - \Theta_{g0}) - (\Theta_{01} - \Theta_{00})$. Independent arm-permutation variances are added before converting them to clipped relative precision,

$$w_{gj} = \text{clip} \left(\frac{1/v_{gj}}{\text{median}_{g,j}(1/v_{gj})}, 0.05, 20 \right).$$

3.2 Structured field estimator

For target-by-coordinate matrix Y , positive weight matrix W , positive-semidefinite hypergraph Laplacian L_H , and nonnegative penalties, the estimator minimizes

$$F_Y(M) = \frac{1}{2} \|Y - M\|_W^2 + \lambda_* \|M\|_* + \lambda_G \text{tr}(M^\top L_H M), \quad (17)$$

where $\|A\|_W^2 = \sum_{g,j} W_{gj} A_{gj}^2$. The normalized hypergraph Laplacian is

$$L_H = I - D_v^{-1/2} H D_e^{-1} W_e H^\top D_v^{-1/2}.$$

Isolated vertices receive zero rows and columns and are unaffected by the graph penalty.

Each covered target formed an unweighted hyperedge with its five highest-cosine covered neighbors. The benchmark used frozen scGPT and gene2vec embeddings; label-permuted scGPT supplied the shuffled-graph control, and targets absent from an embedding were isolated. In PerturbSci, Frangieh, and Papalexi rotations, the other two units selected penalties by symmetric cross-prediction mean squared error. The nuclear penalty was a fraction of the largest training-matrix singular value, on grid $\{0, 0.01, 0.03, 0.1, 0.3, 0.6, 1\}$; the graph grid was $\{0, 0.05, 0.2, 1, 5, 20\}$. Nuclear-only, graph-only, and joint fits used the corresponding grids. The fixed variance-weighted fit used nuclear fraction 0.1 and graph penalty 5; scalar shrinkage used 26 multipliers from 0 to 1.25, and endpoint ridge used $\{10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10, 100\}$. MultiPerturb, PerturbFate, and Arce used the same five-neighbor graph and fixed penalties.

Proposition S5 (Uniqueness and nonexpansive stability). *If $\min_{g,j} W_{gj} > 0$, Eq. (17) has a unique minimizer. For two input matrices Y_1, Y_2 with the same weights, hypergraph, and penalties, their minimizers satisfy*

$$\|\widehat{M}_1 - \widehat{M}_2\|_W \leq \|Y_1 - Y_2\|_W. \quad (18)$$

Proof. The weighted quadratic loss is strongly convex and both penalties are convex, so their sum has a unique minimizer. Let R denote the combined penalty and choose $G_k \in \partial R(\widehat{M}_k)$. From $W \odot (\widehat{M}_k - Y_k) + G_k = 0$, subtracting the two optimality equations and taking the Frobenius inner product with $\widehat{M}_1 - \widehat{M}_2$ gives, by monotonicity of the convex subdifferential,

$$\|\widehat{M}_1 - \widehat{M}_2\|_W^2 \leq \langle W \odot (Y_1 - Y_2), \widehat{M}_1 - \widehat{M}_2 \rangle.$$

Weighted Cauchy-Schwarz proves Eq. (18). $\square$

Accelerated proximal gradient descent soft-thresholds singular values in the proximal step and handles the weighted residual and graph quadratic form in the smooth gradient. Iteration stops at a relative Frobenius step below 10^{-8} ; nonconverged estimates are rejected.

4 Public binary studies

Table 3: Binary RNA-protein studies.

Study	Accession	Linked object	Retained set	Unit and support rule
COMBAT	Zenodo 6120249	RNA-ADT states in one cell	81 ordered pairs	12 Oxford calibration and 24 Oxford pilot samples; development comparison only
BMMC	GSE194122	RNA-ADT states in one cell	100 ordered pairs	two fit donors and four bridge-donor batches; development comparison only
Stephenson	E-MTAB- 10026	RNA-ADT states in one cell	81 ordered pairs	12 Cambridge calibration, 24 Cambridge pilot, and 56 Newcastle held donors
Oh pregnancy cohort	GSE239452	RNA-ADT states in one cell	81 ordered pairs	7 nonpregnant calibration, 8 nonpregnant pilot, and 9 pregnant held donors; post-access corrected replication

Study	Accession	Linked object	Retained set	Unit and support rule
Babdor immuno-microbiome	GSE314416	RNA-ADT states in one cell	81 primary and 576 secondary ordered pairs	12 DB1 calibration and 20 pilot donors from DB1 and DB2; 77 donors from DB3 to DB7 remained unevaluated after the pilot criterion failed
Kelley biliary-cancer trial	GSE317605	RNA-ADT states in one cell	256 ordered pairs	7 calibration patients at four visits; 3 pilot and 8 held patients remained unevaluated after the calibration criterion failed

4.1 COMBAT and BMMC development panels

The COMBAT analysis used the official linked RNA-ADT object and sample manifest [5]. Metadata-only stratification assigned 12 Oxford samples to calibration and 24 to development. Each contributed 512 cells selected deterministically without using assay counts and 81 ordered pairs from nine matched markers.

The BMMC analysis used GSE194122 as distributed with SCMMIB [6]. Donors 11466 and 19593 supplied the fit, and four batches from bridge donor 15078 supplied development evaluation. It included 4,607 cells and 100 ordered pairs from ten matched markers. Evaluation units in both panels participated in model selection.

4.2 Stephenson held-site confirmation

The Stephenson study profiled COVID-19 immune responses across UK sites [7]. Cambridge and Newcastle used BioLegend TotalSeq-C panel 99813; the protocol excluded 11 Sanger/London donors assayed with panel 99814 and six LPS-stimulated Newcastle donors. For each eligible donor, a deterministic rule independent of assay counts selected 512 cells. RNA state was raw UMI count above zero, and within-sample median splitting of each ADT marker fixed its margin at 256/256. Nine matched markers defined 81 ordered pairs, of which at least 64 had to be informative.

The prespecified split contained 12 Cambridge calibration donors (nine COVID-19 and three normal), 24 Cambridge pilot donors (18 COVID-19 and six normal), 11 unused Cambridge donors, and 56 Newcastle held donors. From the grid in Table 1, the pilot selected heterogeneity 0.1, ridge 0.01, graph 0, and $\alpha = 1$; the strongest statistic method was uncentered signed-root deviance with $\alpha = 0.75$. The final source fit used all 36 development donors.

Pilot mean deviance per cell was 0.005079 for the hierarchical estimator and 0.007761 for the selected statistic, a 34.55% reduction (paired donor-bootstrap loss-difference interval -0.003568 to -0.001827 ; 21/24 donors favorable). Relative to destroyed links, the reduction was 71.88% (-0.015650 to -0.009901 ; 23/24 favorable). Prediction then used Newcastle RNA and protein margins alone; paired Newcastle measurements entered once, during evaluation.

4.3 GSE239452 cohort replication

GSE239452 measured pregnancy-specific immunity after SARS-CoV-2 infection [8] and links GSE239450 gene expression with GSE239449 antibody-derived tags. Eligibility criteria excluded donor 100 for fewer than 512 shared cells and donor 78 for missing ADT data. The remaining split comprised seven nonpregnant calibration donors, eight nonpregnant pilot donors, and nine pregnant recipients: three controls, three asymptomatic infections, and three severe infections.

A deterministic rule independent of assay counts selected 512 shared barcodes per donor; state definitions, marker panel, pair set, and support threshold matched Stephenson. Seven-fold calibration selected heterogeneity 1, ridge 0.01, graph 0, and $\alpha = 1$; the comparator was uncentered signed-root deviance with $\alpha = 0.75$. After correction of its inverse reconstruction, pilot mean deviance was 0.008197 versus 0.010247, a 20.01% reduction (-0.004348 to -0.000546 ; 7/8 donors favorable). The destroyed-link reduction was 68.81% (-0.025268 to -0.012083 ; 8/8 favorable). The post-access correction changed only statistic-transfer predictions; primary and destroyed-link predictions and held tables were unchanged. GSE239452 is therefore a post-access corrected cohort replication.

The fixed-interaction Poisson analysis selected $\alpha = 1$ on 15 development donors. Its saturated source fit reconstructed all 81 pooled tables with maximum normalized cell error 1.78×10^{-16} .

4.4 GSE314416 held-pool attempt

GSE314416 profiled healthy participants across seven acquisition pools [9]. The prespecified split assigned 12 DB1 donors to calibration, 20 donors from DB1 and DB2 to the pilot, and 77 donors from DB3 to DB7 to held-pool evaluation. The primary 81-pair panel matched Stephenson; a secondary 24-marker panel contained 576 ordered pairs. Selection independent of assay counts retained 512 cells per donor.

Primary pilot loss was 0.001927, versus 0.001936 for selected signed Pearson-statistic transfer: a 0.49% reduction (paired donor interval -0.0000182 to -0.0000011; 13/20 favorable). The destroyed-link reduction was 0.45% (-0.0000390 to 0.0000223; 11/20 favorable), and the secondary panel reduced statistic-transfer loss by 2.54%. Neither panel reached the fixed 5% threshold, so the 77 held donors were not evaluated.

4.5 GSE317605 longitudinal calibration

GSE317605 profiled longitudinal PBMC RNA and surface proteins during a phase II trial of pembrolizumab plus granulocyte-macrophage colony-stimulating factor in advanced biliary cancer [10]. The split, fixed before assay-value access, assigned seven complete patients to calibration, three complete patients to the pilot, and eight patients to held evaluation. Five held patients had complete visit series and three had partial series. Each calibration visit retained 192 count-blind selected cells and 256 ordered pairs from 16 matched RNA and ADT markers.

For visit v at time $t(v)$ and marker pair e , the longitudinal model decomposed the log odds into a time-specific population field $\beta_{t(v),e}$ and visit deviation u_{ve} . With V_t denoting visits at time t , it minimized

$$Q_{\text{long}} = \sum_t \frac{1}{|V_t|} \sum_{v \in V_t} \left\{ \sum_e \ell_{ve}(\beta_{te} + u_{ve}) + \frac{\eta}{2} \|u_v\|_2^2 \right\} + \frac{0.01}{2} \|\beta\|_F^2 \\ + \frac{\gamma}{2} \sum_t \beta_t^T L_{\text{pair}} \beta_t + \frac{\tau}{2} \sum_{t=1}^3 \|\beta_{t+1} - \beta_t\|_2^2, \quad (19)$$

where ℓ_{ve} is the exact fixed-margin negative log likelihood. The marker operator was $L_{\text{pair}} = L_H \otimes I + I \otimes L_H$, with L_H the normalized Laplacian of a fixed marker hypergraph; temporal smoothing followed the path from T01 to T04. The direct comparator fitted one time-specific interaction per pair by ridge-penalized profiled Poisson likelihood, then profiled row and column terms at each target margin. Both methods used identical patient-disjoint folds and transport grids. Loss was averaged across pairs within each visit, then across visits within each patient, with patients weighted equally.

Leave-one-patient-out calibration crossed donor-deviation penalties $\{0.1, 1, 10\}$, hypergraph and temporal penalties $\{0, 0.05, 0.2\}$, and transport multipliers $\{0, 0.5, 0.75, 1, 1.25\}$. The matched classical comparator crossed ridge penalties $\{0.001, 0.01, 0.1, 1\}$ with the same transport grid. The selected field fit used deviation penalty 0.1, temporal penalty 0.2, hypergraph penalty zero, and transport 0.75. The Poisson fit selected ridge penalty 1 and transport 0.75. Mean deviance was 0.006530, compared with 0.006609 for Poisson prediction. The reduction was 1.19%, with favorable results in 6/7 patients and 4/4 time points. Relative to destroyed links, deviance fell by 17.39%, with favorable results in 7/7 patients. Progression required at least 5% improvement over Poisson and strict improvement over a separately retuned graph-zero fit. The primary and graph-zero fits were identical, so the calibration criterion was not met and the pilot and held assay values were not accessed.

5 Binary conditional transfer results

5.1 Held outcomes

The Stephenson pilot required at least 5% lower donor-equal mean deviance than both statistic-transfer and destroyed-link controls, a 20,000-draw paired donor-bootstrap interval with upper endpoint below zero, and at least 19/24 favorable donors. Held-site confirmation further required at least 45/56 favorable donors and one-sided sign-test $P \leq 0.025$. GSE239452 used corresponding favorable-donor thresholds of 7/8 in pilot and 8/9 in held evaluation; its corrected statistic comparison is descriptive because repair followed outcome access.

Table 4: Held comparison with selected signed-statistic transfer.

Study and evidence role	Hierarchical	Statistic transfer	Reduction	Difference interval	Favorable donors
Stephenson, pre-outcome held site	0.01220	0.01478	17.46%	-0.00413 to -0.00080	50/56; sign $P = 5.09 \times 10^{-10}$
GSE239452, post-access corrected replication	0.00851	0.01413	39.81%	-0.00710 to -0.00423	9/9

Intervals are 20,000-draw paired physical-donor bootstrap intervals for hierarchical-minus-comparator mean deviance per cell.

Both studies selected $\lambda_G = 0$, making the graph-zero diagnostic identical to the selected fit.

Table 5: Held comparison with destroyed links.

Study and evidence role	Hierarchical	Destroyed	Reduction	Difference interval	Favorable donors
Stephenson, pre-outcome held site	0.01220	0.02274	46.36%	-0.01323 to -0.00749	51/56
GSE239452, post-access corrected replication	0.00851	0.04029	78.89%	-0.03567 to -0.02816	9/9

Intervals and loss definitions follow Table 4.

5.2 Stephenson linkage and lineage sensitivity

The prespecified destroyed-link control deterministically permuted complete ADT profiles within each source donor. Of 32 independent post hoc repetitions, 29 met the numerical criterion, and every retained permutation had higher mean held loss than intact linkage. Against the donor-wise mean of retained controls, intact linkage reduced loss by 47.30% (paired donor interval 37.56% to 54.59%; loss-difference interval -0.01367 to -0.00795; 51/56 favorable, $P = 5.85 \times 10^{-11}$). Nested resampling over source permutations and held donors gave a loss-difference interval of -0.01371 to -0.00791.

At a common 64-cell donor budget, the same fixed configurations reduced loss by -3.41% in seven scorable T-cell donors and -0.78% in ten B-cell donors. Among 46 myeloid donors, the reduction was 15.51% (5.51% to 24.61%), with 21/46 favorable ($P = 0.769$). Only cell support changed; the site split and estimator settings remained fixed.

5.3 Fitted interaction baselines

Post hoc fitted-interaction comparisons included a stratified common-effect estimate, pooled-table log odds with conditional reconstruction, and a fixed-interaction Poisson model that profiles recipient marginal parameters.

Table 6: Held comparisons with fitted interaction baselines.

Study	Comparator	Loss, hierarchy; comparator	Relative change (95% CI)	Loss-difference 95% CI	Favorable; exact P
Stephenson	exact common conditional	0.01220; 0.01300	6.18% lower (3.84% to 8.47%)	-0.00121 to -0.00046	46/56; 6.23×10^{-7}
Stephenson	pooled-table log odds, conditional reconstruction	0.01220; 0.01263	3.39% lower (1.42% to 5.37%)	-0.00075 to -0.00017	38/56; 0.00523
GSE239452	exact common conditional	0.00851; 0.00832	2.20% higher (1.34% to 3.03%)	0.00010 to 0.00028	0/9; 1.000
GSE239452	pooled-table log odds, conditional reconstruction	0.00851; 0.00996	14.58% lower (11.94% to 17.16%)	-0.00181 to -0.00115	9/9; 0.00195
GSE239452	fixed-interaction Poisson prediction	0.00851; 0.00998	14.79% lower (12.18% to 17.37%)	-0.00184 to -0.00117	9/9; 0.00195

Relative and loss-difference intervals use 20,000 paired donor-bootstrap draws. Differences are hierarchical-minus-comparator deviance per cell. Exact P values are one-sided binomial sign tests with zero differences counted as nonfavorable. All comparisons are post hoc and descriptive.

5.4 Matched development comparisons

Table 7 compares the conditional estimator with signed-statistic transfer in two adaptive development panels; reported units participated in selection, and intervals are descriptive.

The corresponding destroyed-link losses were 0.046395 in COMBAT and 0.013351 in BMMC, reductions of 75.09% and 18.73%. Both selected graph penalty zero; the best positive-graph configurations were 1.9% and 5.6% worse, respectively.

5.5 Public analysis sequence

Each analysis separated development selection, prediction from recipient margins, and evaluation (Table 8).

Table 7: Adaptive conditional-versus-statistic comparisons.

Panel	Hierarchical	Selected statistic transfer	Reduction	Difference interval	Favorable units
COMBAT, 24 Oxford pilot samples	0.011558	uncentered deviance, 0.015605	25.94%	−0.005967 to −0.002248	17/24 samples
BMMC, 4 bridge-donor batches	0.010851	centered deviance, 0.013148	17.47%	−0.005036 to −0.000385	4/4 batches

Table 8: Public analysis plans and study outcomes.

Study	Planned split	Evaluation status	Result or stopping reason
Stephenson	12 calibration, 24 pilot, 56 held-site donors	Pre-outcome held-site evaluation	Passed both prespecified comparators and donor-level criteria.
GSE239452	7 calibration, 8 pilot, 9 cohort-shift donors	Post-access corrected comparison	Corrected statistic inversion; reported as descriptive replication.
GSE314416	12 calibration, 20 pilot, 77 held-pool donors	Pilot evaluation	Missed the 5% and favorable-donor criteria; held pools were not evaluated.
GSE317605	7 calibration, 3 pilot, 8 held patients	Calibration evaluation	Missed the Poisson and graph-zero criteria; later stages were not evaluated.
GSE179221	Donor-disjoint source, development, and held sets	Eligibility assessment	Cognate feature-axis uniqueness failed before model fitting.
GSE342939	28 source and 6 held donors	Source evaluation	Every tested configuration failed in at least one source fold; held donors were not evaluated.

6 Secondary multistate results

6.1 Data and preprocessing

Table 9: Multistate studies and evaluation units.

Study	Accession	Linked object	Evaluation unit and support
PerturbSci-Kinetics [11]	GSE218566	old and new RNA in one cell	three sequence-distinct guide rotations; 85 targets
Frangieh [12]	scPerturb, Zenodo 13350497	RNA and surface protein in one cell	three guides per target and at least 20 cells per context; 179 targets
Papalexi [13]	scPerturb, Zenodo 13350497	RNA and surface protein in one cell	three deposited replicates and at least 20 cells per target; 24 targets
MultiPerturb [14]	GSE277747	RNA and ATAC in one nucleus	two sequence-distinct guides and at least 30 cells per guide; 35 targets
PerturbFate [15]	GSE291147	pre-existing and nascent RNA in one cell	four technical dates and at least 20 cells in every arm; 49 targets
Arce [16]	GSE278572	cognate RNA-ADT states in one cell	donor A development and donor B evaluation; 28 targets
ReSisTrace [17]	GSE223003	barcode-linked sister lineages	two biological replicates; 186 to 420 lineages per arm

PerturbSci used archived old- and new-RNA states from an encoder fitted to pooled non-targeting controls; guide sequences and controls were divided deterministically into three rotations. Frangieh and Papalexi used harmonized scPerturb RNA and protein objects [18], with separate principal-component and three-state k -means encoders fitted to control cells. MultiPerturb used archived RNA and ATAC states in reciprocal held-guide directions. PerturbFate fitted a three-state encoder to pooled non-targeting controls across 108,114 cells. After Souporecell exclusions, Arce included 100,086 cells and 112 cognate RNA-ADT features; donor A fixed the estimator and donor B supplied evaluation.

ReSisTrace used the 128 highest-variance pretreatment genes and an eight-component principal-component analysis fitted across pretreatment libraries. Three states were fitted to 507 control lineages. Barcode-defined lineages linked pre- and post-treatment states within each condition and replicate; treatment fields subtracted handling-control fields. Intervals used 2,000 bootstrap draws over linked lineages within each deposited replicate.

6.2 Predictive comparisons

Table 10 reports the structured estimator, strongest observed standardized-RMSE baseline, and destroyed-link control for each predictive multistate panel. PerturbSci passed guide replication but not the strongest observed RMSE comparison.

Table 10: Matched multistate comparisons.

Panel	Method	Pearson r (interval)	Standardized RMSE (interval)
PerturbSci	structured field	0.525 (0.483, 0.566)	0.844 (0.819, 0.868)
	marginal + structured residual	0.545 (0.507, 0.584)	0.831 (0.801, 0.862)
	destroyed link	0.074 (0.003, 0.144)	1.106 (1.071, 1.141)
Frangieh	structured field	0.040 (0.013, 0.066)	1.009 (1.002, 1.017)
	scalar shrinkage	0.007 (−0.026, 0.041)	0.999 (0.998, 1.001)
	destroyed link	0.092 (0.064, 0.122)	1.187 (1.164, 1.210)
Papalexi	structured field	−0.025 (−0.150, 0.096)	1.037 (1.006, 1.076)
	zero field	NA	1.000 (1.000, 1.000)
	destroyed link	−0.044 (−0.158, 0.058)	1.223 (1.139, 1.317)
MultiPerturb	structured field	−0.224 (−0.330, −0.124)	1.047 (1.005, 1.095)
	zero field	NA	1.000 (1.000, 1.000)
	destroyed link	−0.011 (−0.065, 0.059)	1.218 (1.145, 1.299)
PerturbFate	structured field	−0.092 (−0.152, −0.035)	1.023 (1.010, 1.035)
	linear cross-covariance	−0.011 (−0.122, 0.092)	1.000 (0.995, 1.004)
	destroyed link	−0.011 (−0.078, 0.056)	1.125 (1.089, 1.167)
Arce donor B	structured field	0.412 (0.331, 0.494)	1.093 (1.012, 1.185)
	linear cross-covariance	0.304 (0.195, 0.404)	0.993 (0.989, 0.996)
	destroyed link	0.170 (0.049, 0.299)	1.194 (1.081, 1.317)

Parentheses give 95% target-resampling intervals.

All eight ReSisTrace condition-by-replicate arms had fixed-margin Monte Carlo randomization $P = 1/65$ and Benjamini-Hochberg (BH) $q = 0.0154$. Each observed statistic and 64 disjoint test permutations was centered against the same independent 64-permutation reference mean. No treatment contrast had a positive lineage-bootstrap lower limit.

Table 11: ReSisTrace treatment-minus-control replication.

Treatment	Replicate cosine	2,000-draw lineage-bootstrap 95% interval	Link P	BH q
Carboplatin	−0.674	−0.965, 0.729	0.877	0.877
Olaparib	0.223	−0.876, 0.925	0.354	0.531
NK	0.952	−0.801, 0.968	0.015	0.046

6.3 PerturbSci sensitivity and functional analysis

Table 12: PerturbSci performance in each held guide rotation.

Held rotation	Pooled Pearson r	Mean target cosine	Standardized RMSE
0	0.506	0.539	0.857
1	0.540	0.579	0.843
2	0.544	0.544	0.830

Table 13: PerturbSci sensitivity of pooled held-guide correlation.

Parameter	Value	Pooled r
States per assay	2	-0.121
	3	0.525
	4	0.640
Reference permutations	16	0.494
	32	0.508
	64	0.525
	128	0.523
Pseudocount	0.25	0.339
	0.50	0.525
	1.00	0.718

Module analysis required each target to enter the top field-norm quintile in every held truth and prediction, excluding targets that also passed after link destruction. No target qualified, so pathway and complex enrichment were not tested; XRN2 passed only in the destroyed control.

Mean top-five cosine-neighbor recovery was 0.064 (95% target-resampling interval 0.050 to 0.079) for the structured fit and 0.051 (0.041 to 0.063) after link destruction. The paired difference was 0.013 (-0.003 to 0.030 ; 10,000-draw target-label permutation $P = 0.283$). Reactome neighbor-edge enrichment failed its controls. CORUM edge enrichment exceeded destroyed links by 0.018 (0.008 to 0.029) and was nominal against random labels ($P = 0.0269$), but not after correction across two annotation families ($q = 0.0538$). No module, neighborhood, or individual edge met all criteria.

7 Limitations

The 81 within-donor pair tables reuse cells and marker vectors, but the composite objective does not model their joint covariance. Held-donor bootstraps condition on the fitted source field, selected configuration, and deterministic cell subsamples. Within-donor median ranks fix every Stephenson protein margin at 256/256, so the held experiment does not evaluate natural protein thresholds, alternative quantiles, or protein-margin shifts. Each primary destroyed-link comparison uses one deterministic within-donor permutation; the Stephenson sensitivity retained 29 of 32 repeated fits, and three did not meet the convergence criterion. In simulation, 16 entity tables are independent, the recipient field has no random effect around the population field, and margin shift is deliberately strong. These simulations separate conditioning from shrinkage under a fixed-margin model without defining a multivariate cell-level generative law.

Conditional likelihood removes row and column nuisance parameters but can lose finite-sample information about the odds ratio when one or both margins are random under the sampling design. The lost fraction tends to zero under standard large-sample conditions [19]. We did not estimate its magnitude in the sparse marker tables used here.

The original GSE239452 statistic inversion mishandled negative values near the lower feasible endpoint, producing 80 coordinate mismatches among 729 reconstructed tables. The corrected inverse produced none at tolerance 10^{-8} and maximum absolute coordinate error below 4.0×10^{-12} . It recomputed only statistic-transfer predictions; primary and destroyed-link predictions and held tables were unchanged. Because repair followed outcome access, GSE239452 is a post-access corrected replication, not a second pre-outcome confirmation.

8 Implementation and reproducibility

The v2.0.3 release at <https://github.com/sushaan-k/coupling-fields-benchmark/releases/tag/coupling-fields-v2.0.3-public-benchmark> contains the MIT-licensed estimator, analysis protocols, cohort and simulation results, complete study record, and method grids. Its verification tests reproduce the reported tables and figures from the derived results. Raw matrices remain available from the accessions in Tables 3 and 9 under their source licenses and are not redistributed.

References

- [1] Liao JG. A hierarchical Bayesian model for combining multiple 2×2 tables using conditional likelihoods. *Biometrics*. 1999;55(1):268–272. <https://doi.org/10.1111/j.0006-341x.1999.00268.x>.
- [2] Coull BA, Agresti A. Generalized log-linear models with random effects, with application to smoothing contingency tables. *Statistical Modelling*. 2003;3(4):251–271. <https://doi.org/10.1191/1471082X03st059oa>.

- [3] Stijnen T, Hamza TH, Özdemir P. Random effects meta-analysis of event outcome in the framework of the generalized linear mixed model with applications in sparse data. *Statistics in Medicine*. 2010;29(29):3046–3067. <https://doi.org/10.1002/sim.4040>.
- [4] Paule RC, Mandel J. Consensus values and weighting factors. *Journal of Research of the National Bureau of Standards*. 1982;87(5):377–385. <https://doi.org/10.6028/jres.087.022>.
- [5] COVID-19 Multi-omics Blood Atlas (COMBAT) Consortium. A blood atlas of COVID-19 defines hallmarks of disease severity and specificity. *Cell*. 2022;185(5):916–938.e58. <https://doi.org/10.1016/j.cell.2022.01.012>.
- [6] Luecken MD, Burkhardt DB, Cannoodt R, Lance C, Agrawal A, Aliee H, et al. A sandbox for prediction and integration of DNA, RNA, and proteins in single cells. In: *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*. vol. 1; 2021. p. 1–13. Available from: <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/158f3069a435b314a80bdc024f8e422-Abstract-round2.html>.
- [7] Stephenson E, Reynolds G, Botting RA, Calero-Nieto FJ, Morgan MD, Tuong ZK, et al. Single-cell multi-omics analysis of the immune response in COVID-19. *Nature Medicine*. 2021;27:904–916. <https://doi.org/10.1038/s41591-021-01329-2>.
- [8] Oh DS, Kim E, Normand R, Lu G, Shook LL, Lyall A, et al. SARS-CoV-2 infection elucidates features of pregnancy-specific immunity. *Cell Reports*. 2024;43(11):114933. <https://doi.org/10.1016/j.celrep.2024.114933>.
- [9] Babdor J, Patel RK, Davidson B, Koser K, Noecker C, Rahim MK, et al. Immune-microbiome coordination defines interferon setpoints in healthy humans. *Cell*. 2026;189(10):3144–3163.e20. <https://doi.org/10.1016/j.cell.2026.02.003>.
- [10] Kelley RK, Bracci P, Gordan JD, Behr SC, Quandt Z, Atreya CE, et al. A Phase II Trial of Pembrolizumab plus Granulocyte-Macrophage Colony-Stimulating Factor in Advanced Biliary Cancer. *Clinical Cancer Research*. 2026;Published online 14 August 2026. <https://doi.org/10.1158/1078-0432.CCR-25-3987>.
- [11] Xu Z, et al. Dissecting key regulators of transcriptome kinetics through scalable single-cell RNA profiling of pooled CRISPR screens. *Nature Biotechnology*. 2024;42:1218–1223. <https://doi.org/10.1038/s41587-023-01948-9>.
- [12] Frangieh CJ, Melms JC, Thakore PI, Geiger-Schuller KR, Ho P, Luoma AM, et al. Multimodal pooled Perturb-CITE-seq screens in patient models define mechanisms of cancer immune evasion. *Nature Genetics*. 2021;53:332–341. <https://doi.org/10.1038/s41588-021-00779-1>.
- [13] Papalexi E, Mimitou EP, Butler AW, Foster S, Bracken B, Mauck WM, et al. Characterizing the molecular regulation of inhibitory immune checkpoints with multimodal single-cell screens. *Nature Genetics*. 2021;53:322–331. <https://doi.org/10.1038/s41588-021-00778-2>.
- [14] Yan RE, et al. Pooled CRISPR screens with joint single-nucleus chromatin accessibility and transcriptome profiling. *Nature Biotechnology*. 2025;43:1628–1634. <https://doi.org/10.1038/s41587-024-02475-x>.
- [15] Xu S, et al. Mapping convergent regulators of melanoma drug resistance by PerturbFate. *Nature*. 2026;654:261–271. <https://doi.org/10.1038/s41586-026-10367-0>.
- [16] Arce MM, et al. Central control of dynamic gene circuits governs T cell rest and activation. *Nature*. 2025;637:930–939. <https://doi.org/10.1038/s41586-024-08314-y>.
- [17] Dai J, et al. Tracing back primed resistance in cancer via sister cells. *Nature Communications*. 2024;15:1158. <https://doi.org/10.1038/s41467-024-45478-7>.
- [18] Peidli S, Green TD, Shen C, Gross T, Min KH, Garda S, et al. scPerturb: harmonized single-cell perturbation data. *Nature Methods*. 2024;21:531–540. <https://doi.org/10.1038/s41592-023-02144-y>.
- [19] Schervish MJ, Kadane JB, Seidenfeld T, Stern RB. Information lost by conditioning on the random margins in a two-by-two table. *Biometrika*. 2025;112(4):asaf064. <https://doi.org/10.1093/biomet/asaf064>.